nature portfolio

Peer Review File

Biofilm formation and dynamics in the marine cyanobacterium Prochlorococcus

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This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have answered all comments by the reviewers and conducted several additional experiments that nicely cover the questions raised. Please check the color coding of lines in figure 4B.

Reviewer #2

(Remarks to the Author)
I believe that the authors have provided a thorough response to the initial review. The additional laboratory-experiments significantly improve the manuscript. Specifically, the addition of direct total dissolved organic carbon measurements provides essential evidence for the claims regarding planktonic and biofilm carbon exudation rates. Furthermore, the implementation of new parallel tracking controls for the emergence experiments, both from the glass culture vessels and the chitin beads, substantially improves the validity of the emergence rate estimations.
I have one concern about the normalization of net oxygen production rate, carbon exudation rate, and chlorophyll content against flow cytometry particle number rather than total biovolume (integrated forward scatter). I appreciate the context regarding the red fluorescence vs forward scatter data and the observation of the elongated tail in the biofilm populations, which the authors hypothesize may represent cell aggregates making up less than 10% of the population. However, an elongated tail on a log-log scatter plot can contribute a perhaps surprisingly large fraction of the total culture biovolume. I recommend a supplementary figure or table showing the equivalent of Figure 2A, D, and F when normalized to total biovolume. Alternatively, including the FSC vs. Red Fluorescence scatter plots in the supplement would allow the reader to visually assess the impact of this population tail on the total biomass.
Minor edits:
• Line 276: “where we could one term” change to “where we could make one term”
• Line 344: “Supp. Fig. 5” change to “Supp. Fig. 6C”.
• Line 729: Figure 4B: The legend is difficult to understand. Perhaps the “Observed Planktonic” should be orange.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)
The authors have done a great job addressing all of my remaining concerns.

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Detailed reviewer comments and responses:

For clarity, we separate multi-part reviews to respond to each question individually. Reviewer comments are in bold and italicized to distinguish them from our responses.

Reviewer #1:

Major comments:

1. The authors have answered all comments by the reviewers and conducted several additional experiments that nicely cover the questions raised. Please check the color coding of lines in figure 4B.

We apologize for the confusion with the legend. The “observed planktonic” box should be orange with a black outline, and the “adherent” box should be purple with a black outline. We have made these changes.

Reviewer #2.

Major comments:

1. I believe that the authors have provided a thorough response to the initial review. The additional laboratory experiments significantly improve the manuscript. Specifically, the addition of direct total dissolved organic carbon measurements provides essential evidence for the claims regarding planktonic and biofilm carbon exudation rates. Furthermore, the implementation of new parallel tracking controls for the emergence experiments, both from the glass culture vessels and the chitin beads, substantially improves the validity of the emergence rate estimations.
I have one concern about the normalization of net oxygen production rate, carbon exudation rate, and chlorophyll content against flow cytometry particle number rather than total biovolume (integrated forward scatter). I appreciate the context regarding the red fluorescence vs forward scatter data and the observation of the elongated tail in the biofilm populations, which the authors hypothesize may represent cell aggregates making up less than 10% of the population. However, an elongated tail on a log-log scatter plot can contribute a perhaps surprisingly large fraction of the total culture biovolume. I recommend a supplementary figure or table showing the equivalent of Figure 2A, D, and F when normalized to total biovolume. Alternatively, including the FSC vs. Red Fluorescence scatter plots in the supplement would allow the reader to visually assess the impact of this population tail on the total biomass.

We have added representative flow cytograms for parental, planktonic, and biofilm populations as Supp. Fig. 4A. We have also added the requested panels as Supp. Fig. 4B-C with the previous single panel Supp. Fig. 4 becoming panel 4E. Normalizing by integrated forward scatter as a proxy for total population biovolume recapitulated the same trends as normalizing by cell count.

We have also added a note about normalization to biovolume at lines 161-165 and references to the new supplementary panels at lines 165, 181-182, and 191. We hope that these new presentations of the data are satisfactory.

Minor comments:

1. Line 276: "where we could one term" change to "where we could make one term"

We have made this change.

2. Line 344: "Supp. Fig. 5" change to "Supp. Fig. 6C".

Thank you for catching this. We have made this edit.

3. Line 729: Figure 4B: The legend is difficult to understand. Perhaps the "Observed Planktonic" should be orange.

We apologize for the confusion with the legend. As you suggest, the "observed planktonic" box should be orange with a black outline; in addition, the "adherent" box should be purple with a black outline. We have made these changes.